Assignment 5

Layout Optimization for LCoE

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1. Task 1: Wind Rose at Hub Height

Wind speed and direction data were downloaded from the New European Wind Atlas (NEWA) for Marienplatz, Munich, covering the full mesoscale time series from 1989 to 2022. The raw data were provided at 10 m height and scaled to hub height (120 m) using the power-law wind shear model with exponent $\alpha = 0.20$:

$$U(h) = U_{\mathrm{ref}} \left(\frac{h}{h_{\mathrm{ref}}} \right)^{\alpha}$$

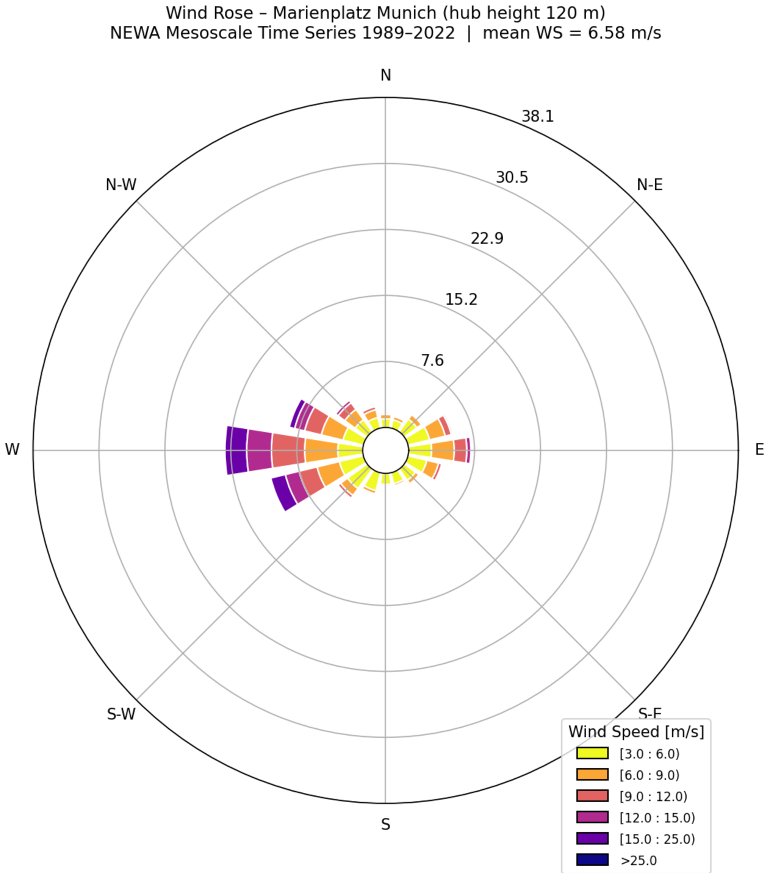

Figure 1: Wind rose at Marienplatz, Munich – NEWA Mesoscale Time Series 1989–2022 (hub height 120 m, mean wind speed = 6.58 m/s).

The wind rose shows a strongly dominant westerly and south-westerly wind regime. Wind speeds of 6–9 m/s (orange) and 9–12 m/s (salmon) are most frequent, and higher speeds above 15 m/s (purple) also arrive predominantly from the west. Wind activity from the north and south is negligible. The mean wind speed at hub height is **6.58 m/s**.

This strongly directional resource is a critical input for the layout optimisation in Task 3: turbines aligned in the west–east direction will experience the largest wake losses under the prevailing wind.

2. Task 2: Baseline Wind Farm Evaluation

2.1 Setup

The initial layout places eight IEA 3.4 MW turbines in a 4×2 grid centred within the 2.8 km × 2.8 km boundary, with the substation at the centre point. Wake interactions were evaluated with the Gauss wake model (default parameters) in FLORIS, using a wind shear exponent of $\alpha = 0.20$ and turbulence intensity TI = 6 %.

2.2 Results

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AEP:                     68.36  GWh/yr
Capacity Factor:         28.95  %
Farm Efficiency:         77.62  %
LCoE:                    62.40  $/MWh
Profitability Index:      0.119  --
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Because the turbines are aligned east–west and the dominant winds arrive from the west, the rear row suffers substantial wake losses, reducing AEP and farm efficiency. The baseline LCoE of 62.40 $/MWh is below the electricity price of 67 $/MWh, so the project is marginally profitable, but there is clear room for improvement through layout optimisation.

3. Task 3: COBYLA Layout Optimisation

3.1 Method

`scipy.optimize.minimize` with the **COBYLA** (Constrained Optimization BY Linear Approximation) gradient-free method was used to optimise the 16 turbine coordinates ($8 \times x$, $8 \times y$) so as to minimise the LCoE of the wind farm. The following constraints were enforced:

- All turbines must remain within the 2.8 km × 2.8 km boundary.
- Minimum turbine-to-turbine separation of $2D = 260$ m for all pairs.

3.2 Convergence

The optimiser starts at approximately 63 $/MWh and converges rapidly; the objective is essentially flat after iteration ∼500, confirming that a local optimum has been reached. The total number of function evaluations was 1 900.

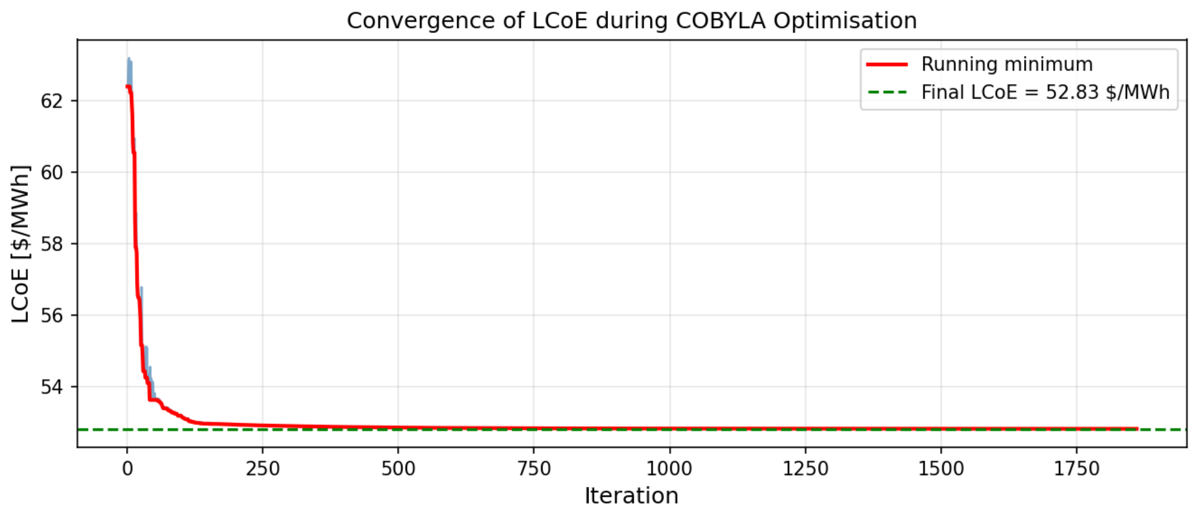

Figure 2: Convergence of LCoE during COBYLA optimisation. The running minimum (red) drops rapidly within the first ~250 iterations and flattens at the final optimum of 52.83 $/MWh (green dashed line).

3.3 Optimised Layout

The optimised layout shows several key changes relative to the baseline:

- Turbines aligned west–east in the baseline have been spread in the north–south direction, reducing wake overlap under the dominant westerly wind.
- Several turbines moved towards the boundary edges to maximise mutual separation.
- All minimum spacing constraints ($2D = 260\,\mathrm{m}$) are satisfied, as confirmed by the non-overlapping exclusion circles.

3.4 Comparison of Results

Parameter	Initial	Optimised	Unit
AEP	68.36	87.28	GWh/yr
Capacity Factor	28.95	36.96	%
Farm Efficiency	77.62	99.11	%
LCoE	62.40	52.83	$/MWh
ΔLCoE	9.57		$/MWh
Profitability Index	0.119	0.462	–

3.5 Profitability Discussion

The optimised LCoE of **52.83 $/MWh** is well below the electricity price of 67 $/MWh, confirming that the project has become clearly profitable after optimisation. The ΔLCoE of 9.57 $/MWh was achieved purely by repositioning turbines, with no additional capital expenditure.

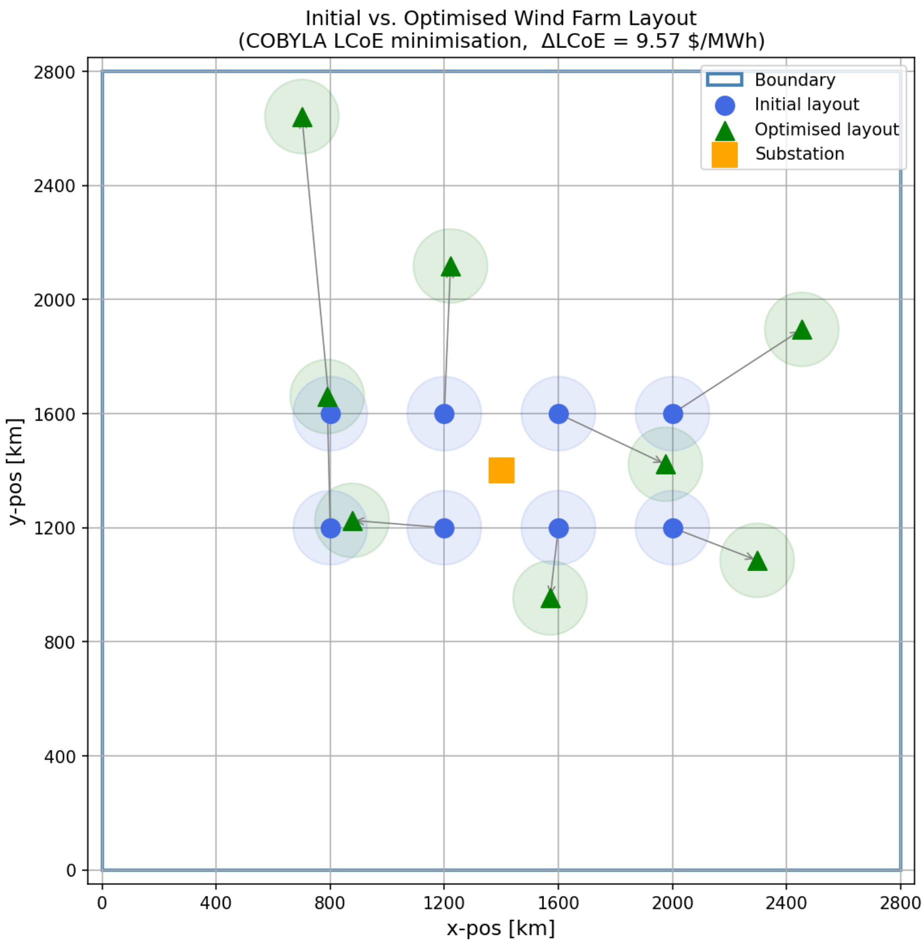

Figure 3: Initial (blue circles) vs. optimised (green triangles) wind farm layout. Arrows indicate turbine displacement. ΔLCoE = 9.57 \$/MWh.

The farm efficiency improvement from 77.62 % to 99.11 % demonstrates that wake losses were substantially reduced, driving up AEP and reducing LCoE. This underlines layout optimisation as a highly cost-effective measure – a finding consistent with the strongly directional wind resource observed at the Munich site.